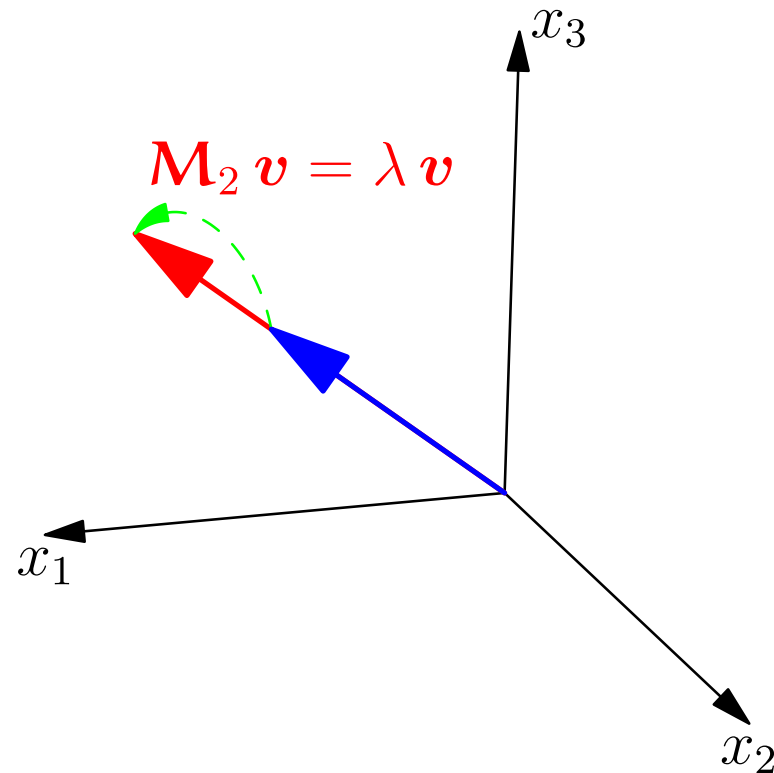


# Mathematics of Machine Learning

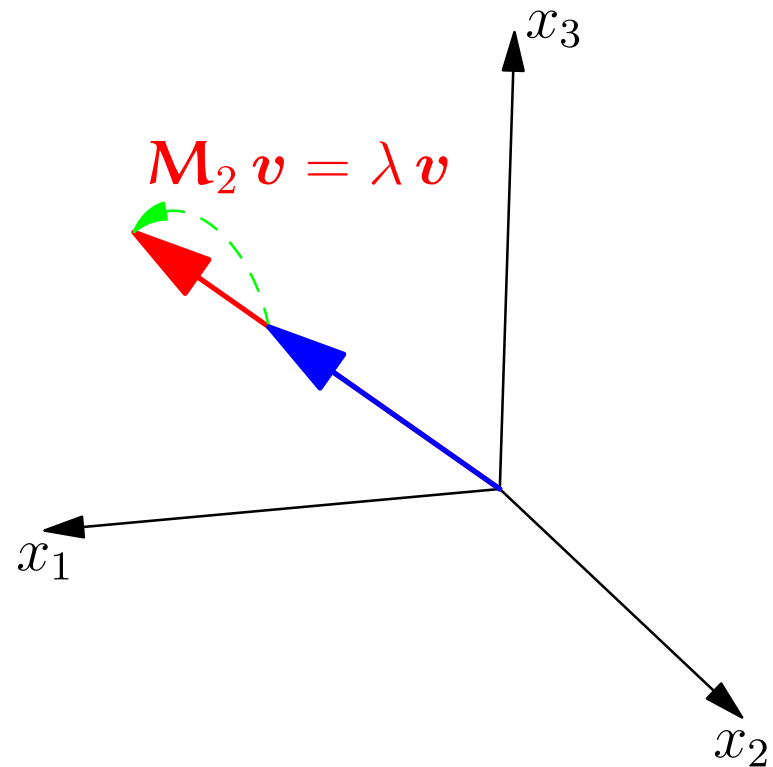
## *Eigensystems*



*Eigenvectors, Orthogonal Matrices, Eigenvector Decomposition, Rank*

# Outline

1. **Eigenvectors**
2. Orthogonal Matrices
3. Eigen Decomposition
4. Low Rank Approximation
5. Non-Normal Matrices



# Eigenvector equation

- Eigen-systems help us to understand mappings
- A non-zero vector  $v$  is said to be an **eigenvector** if

$$Mv = \lambda v$$

- $M$  is square (i.e.  $n \times n$ )
- Where the number  $\lambda$  is the **eigenvalue**
- Eigenvalues play a fundamental role in understanding operators

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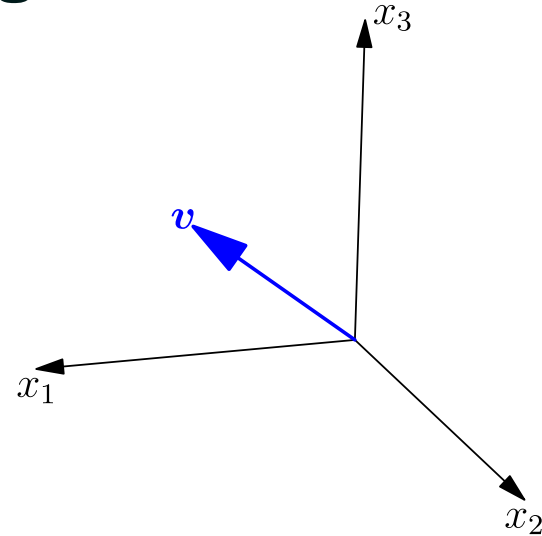
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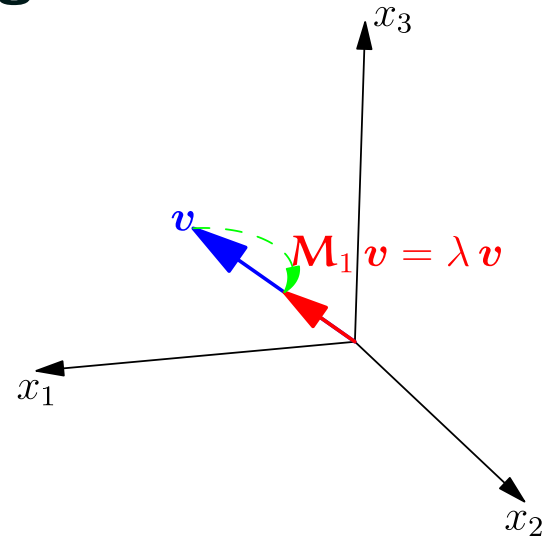


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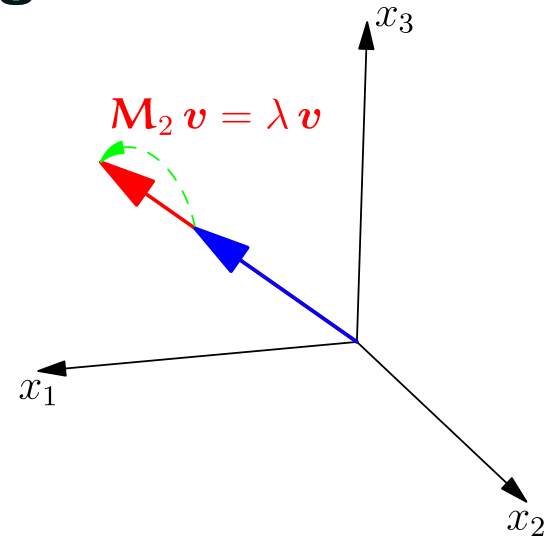


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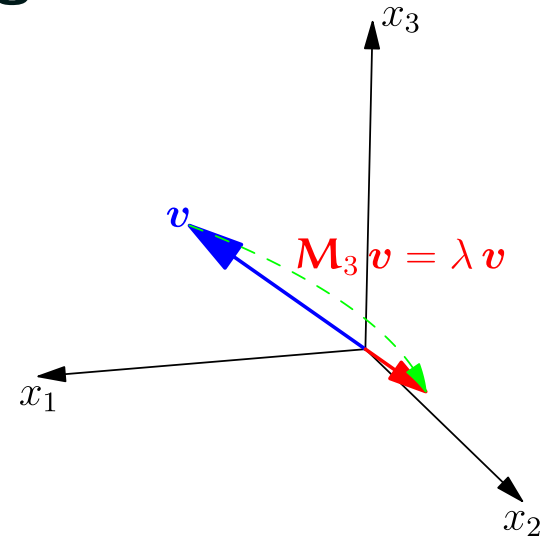


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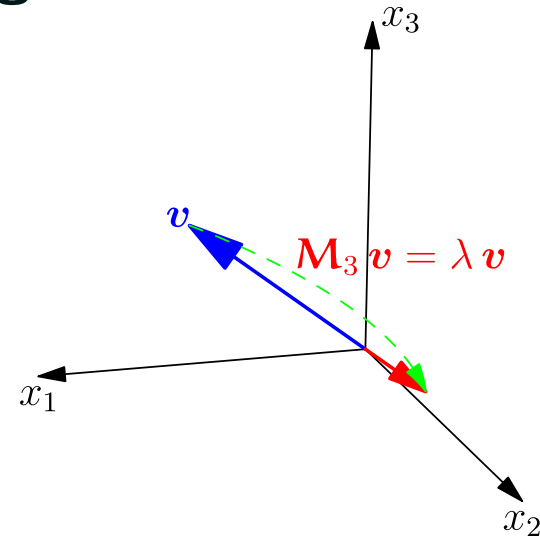


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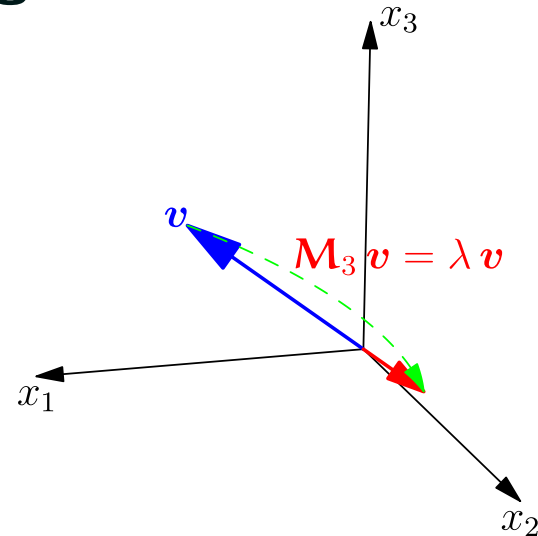


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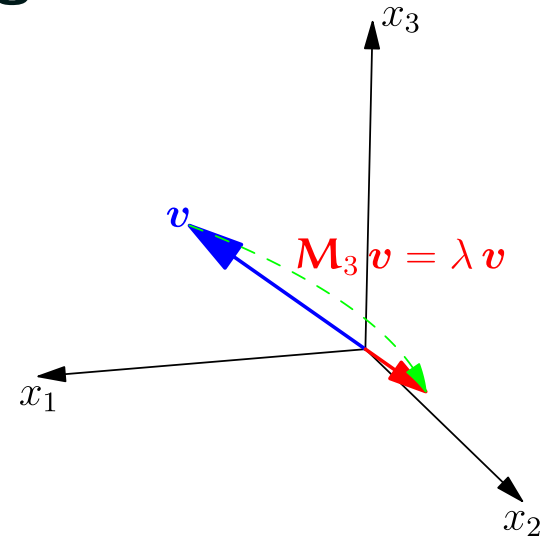


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# Symmetric Matrices

- If  $\mathbf{M}$  is an  $n \times n$  **symmetric** matrix then it has  $n$  real orthogonal eigenvectors with real eigenvalues
- We denote the  $i^{th}$  eigenvector by  $\mathbf{v}_i$  and the corresponding eigenvalue by  $\lambda_i$  so that

$$\mathbf{M}\mathbf{v}_i = \lambda_i \mathbf{v}_i$$

- Orthogonal means that if  $i \neq j$  then

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# Proof of Orthogonality

- $(\mathbf{M}\mathbf{v}_i = \lambda_i\mathbf{v}_i)^\top$  implies  $\mathbf{v}_i^\top \mathbf{M}^\top = \lambda_i\mathbf{v}_i^\top$
- When  $\mathbf{M}$  is symmetric then  $\mathbf{M}\mathbf{v}_i = \lambda_i\mathbf{v}_i \Rightarrow \mathbf{v}_i^\top \mathbf{M} = \lambda_i\mathbf{v}_i^\top$
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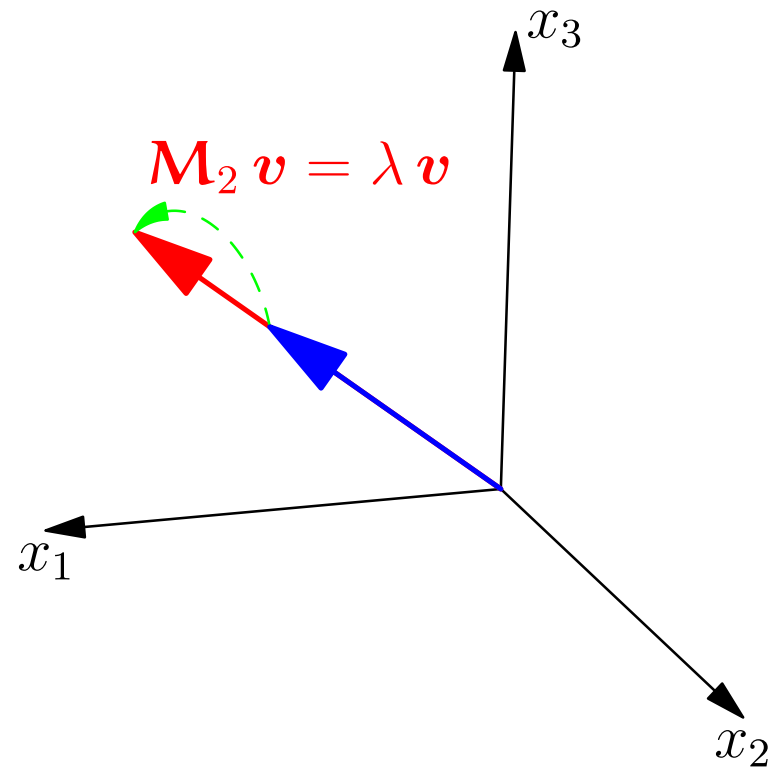
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# Outline

1. Eigenvectors
2. **Orthogonal Matrices**
3. Eigen Decomposition
4. Low Rank Approximation
5. Non-Normal Matrices



# Orthogonal Matrices

- We can construct an **orthogonal** matrix **V** from the normalised eigenvectors ( $\mathbf{v}_i^T \mathbf{v}_i = 1$ )

$$\mathbf{V} = (\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n)$$

- Matrix **V** is an  $n \times n$  matrix
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- Thus multiply both sides on the left by  $\mathbf{V}$

$$\mathbf{V} \mathbf{V}^T \mathbf{V} = \mathbf{V}$$

- $\mathbf{V}$  will have an inverse (we show this on the next slide),  $\mathbf{V}^{-1}$ , such that  $\mathbf{V} \mathbf{V}^{-1} = \mathbf{I}$
- Multiplying the equation on the right by  $\mathbf{V}^{-1}$

$$(\mathbf{V} \mathbf{V}^T) \mathbf{V} \mathbf{V}^{-1} = \mathbf{V} \mathbf{V}^{-1}$$

$$\mathbf{V} \mathbf{V}^T = \mathbf{I}$$

- Note that,  $\mathbf{V}^{-1} = \mathbf{V}^T$  (definition of orthogonal matrix)

# Invertible Matrices

- A matrix,  $\mathbf{M}$ , will be singular (not invertible) if there exists a vector  $\mathbf{x}$  ( $\neq \mathbf{0}$ ) such that

$$\mathbf{M}\mathbf{x} = \mathbf{0}$$

- Now if there exists such a vector such that  $\mathbf{V}\mathbf{x} = \mathbf{0}$  then multiplying by  $\mathbf{V}^T$  we get

$$\mathbf{V}^T\mathbf{V}\mathbf{x} = \mathbf{V}^T\mathbf{0}$$

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- Thus  $\mathbf{V}$  is invertible

# Rotations

- Orthogonal matrices satisfy  $\mathbf{V}^\top \mathbf{V} = \mathbf{V} \mathbf{V}^\top = \mathbf{I}$
- As a consequence they define rotations (and possibly a reflection)
- Consider a vector  $\mathbf{x}$  and  $\mathbf{x}' = \mathbf{V}\mathbf{x}$ , now

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- Rotations and reflections preserve lengths and angles

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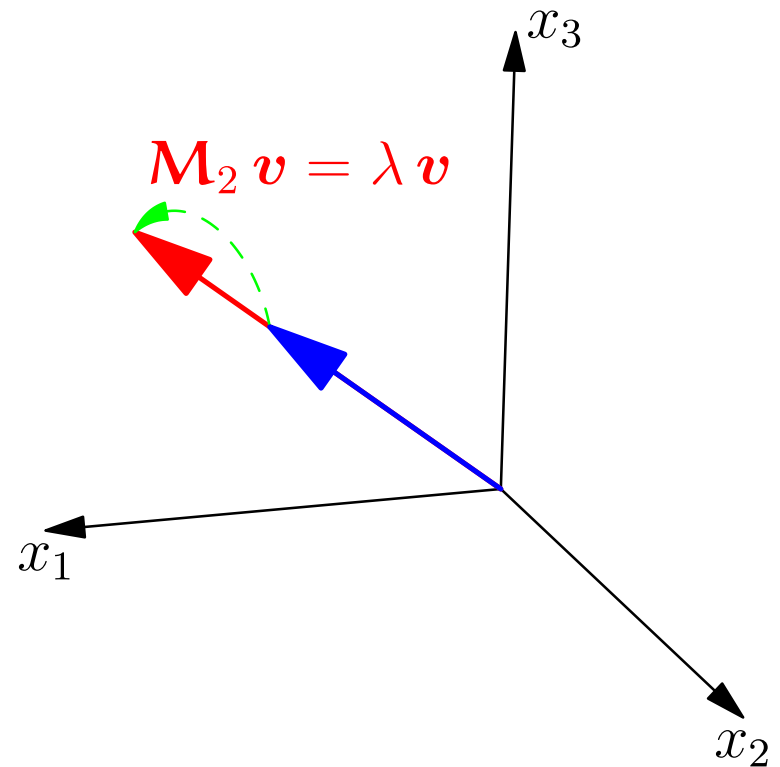
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# Outline

1. Eigenvectors
2. Orthogonal Matrices
3. **Eigen Decomposition**
4. Low Rank Approximation
5. Non-Normal Matrices



# Matrix Decomposition

- Taking the matrix of eigenvectors,  $\mathbf{V}$ , then

$$\mathbf{M}\mathbf{V} = \mathbf{M}(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n) = (\lambda_1 \mathbf{v}_1, \lambda_2 \mathbf{v}_2, \dots, \lambda_n \mathbf{v}_n) = \mathbf{V}\mathbf{\Lambda}$$

- where  $\mathbf{\Lambda} = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_n) = \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{pmatrix}$

- Now

$$\mathbf{M} = \mathbf{M}\mathbf{V}\mathbf{V}^T = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^T$$

- Very important *similarity transform*

# Matrix Decomposition

- Taking the matrix of eigenvectors,  $V$ , then

$$MV = M(v_1, v_2, \dots, v_n) = (\lambda_1 v_1, \lambda_2 v_2, \dots, \lambda_n v_n) = V\Lambda$$

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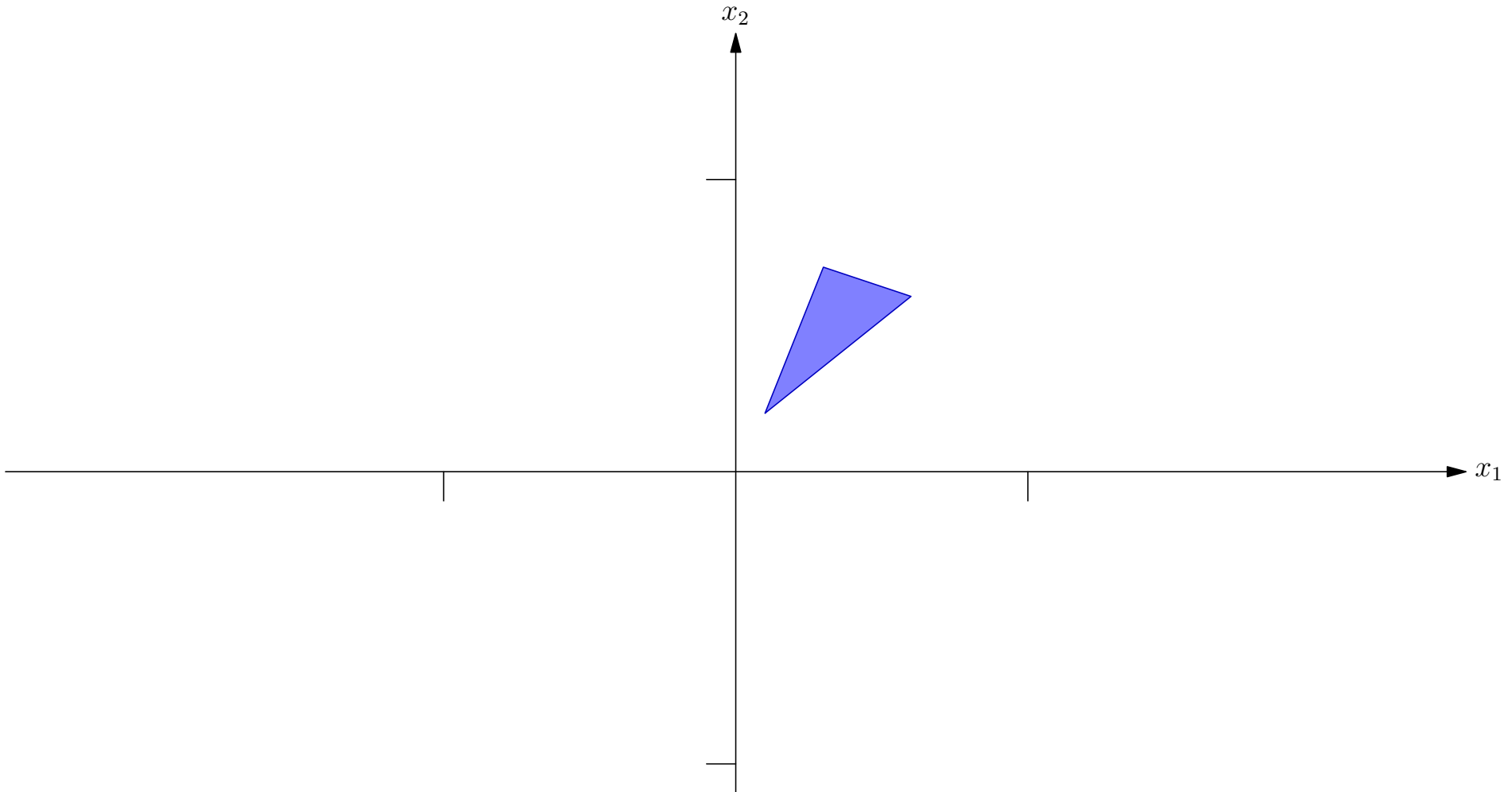
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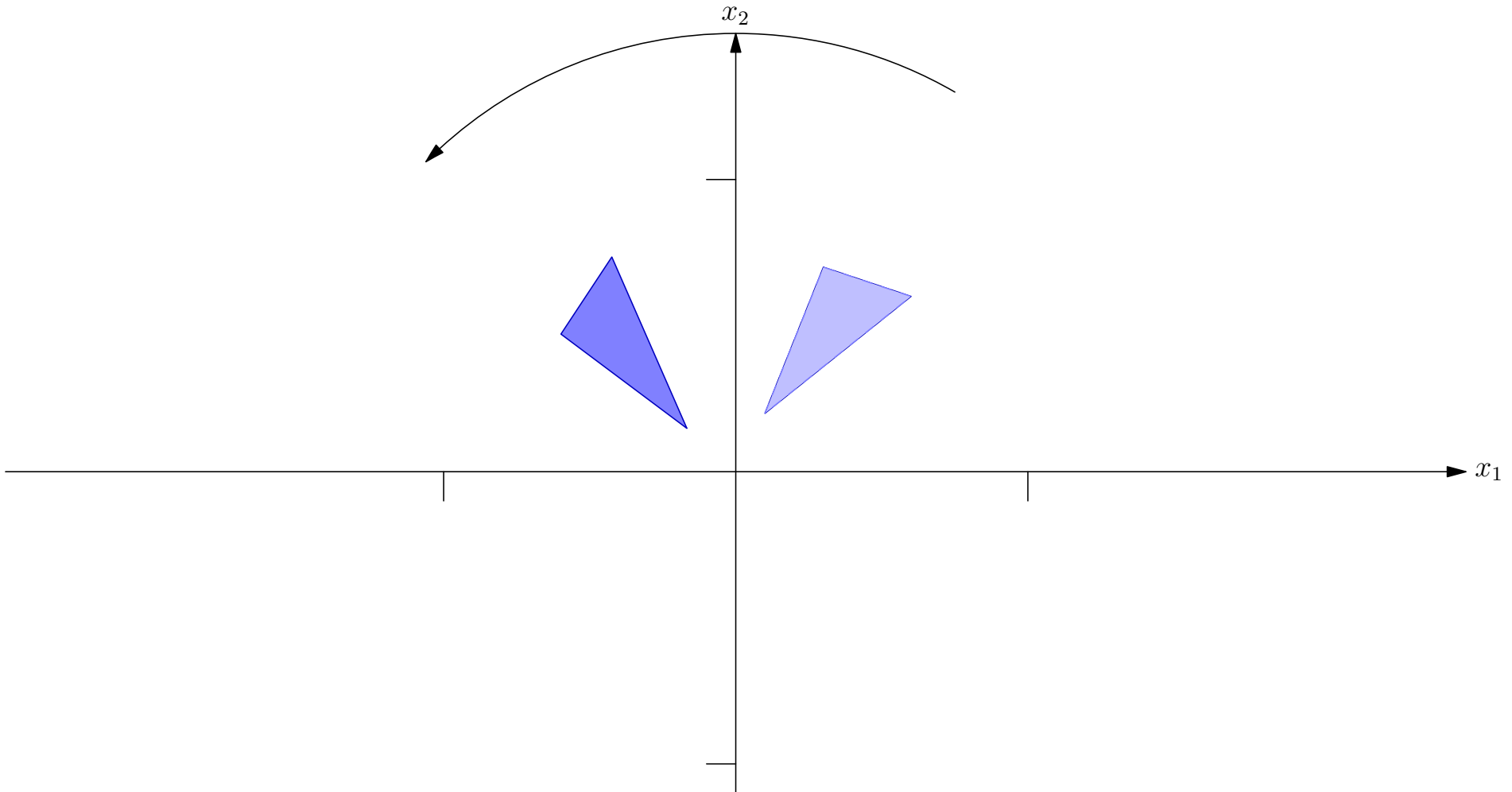
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$$\mathbf{M} = \begin{pmatrix} 0.83 & -0.31 \\ -0.31 & 1.9 \end{pmatrix} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^T = \begin{pmatrix} \cos(-75) & \sin(-75) \\ -\sin(-75) & \cos(-75) \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 0 & 0.75 \end{pmatrix} \begin{pmatrix} \cos(75) & \sin(75) \\ -\sin(75) & \cos(75) \end{pmatrix}$$



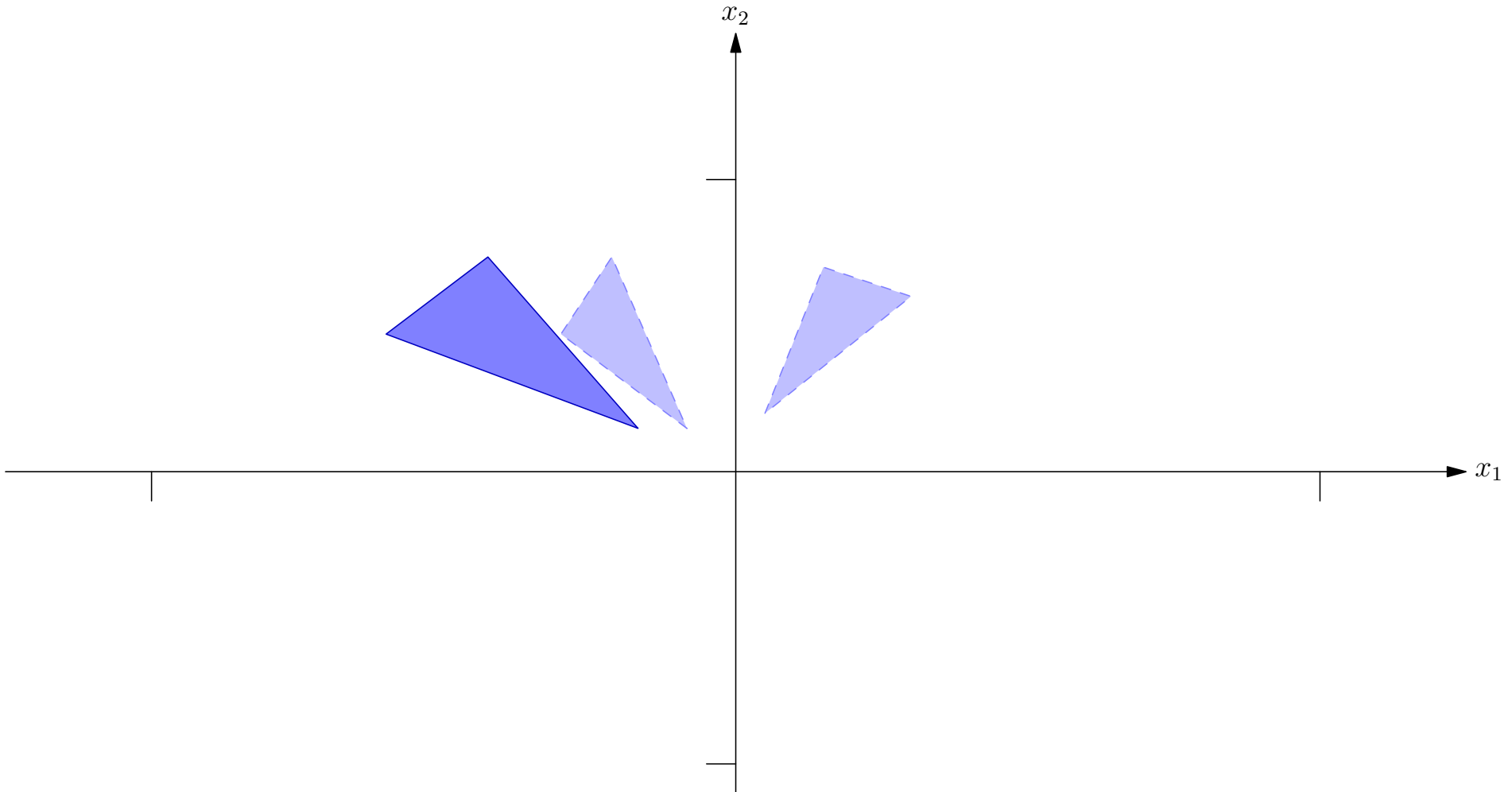
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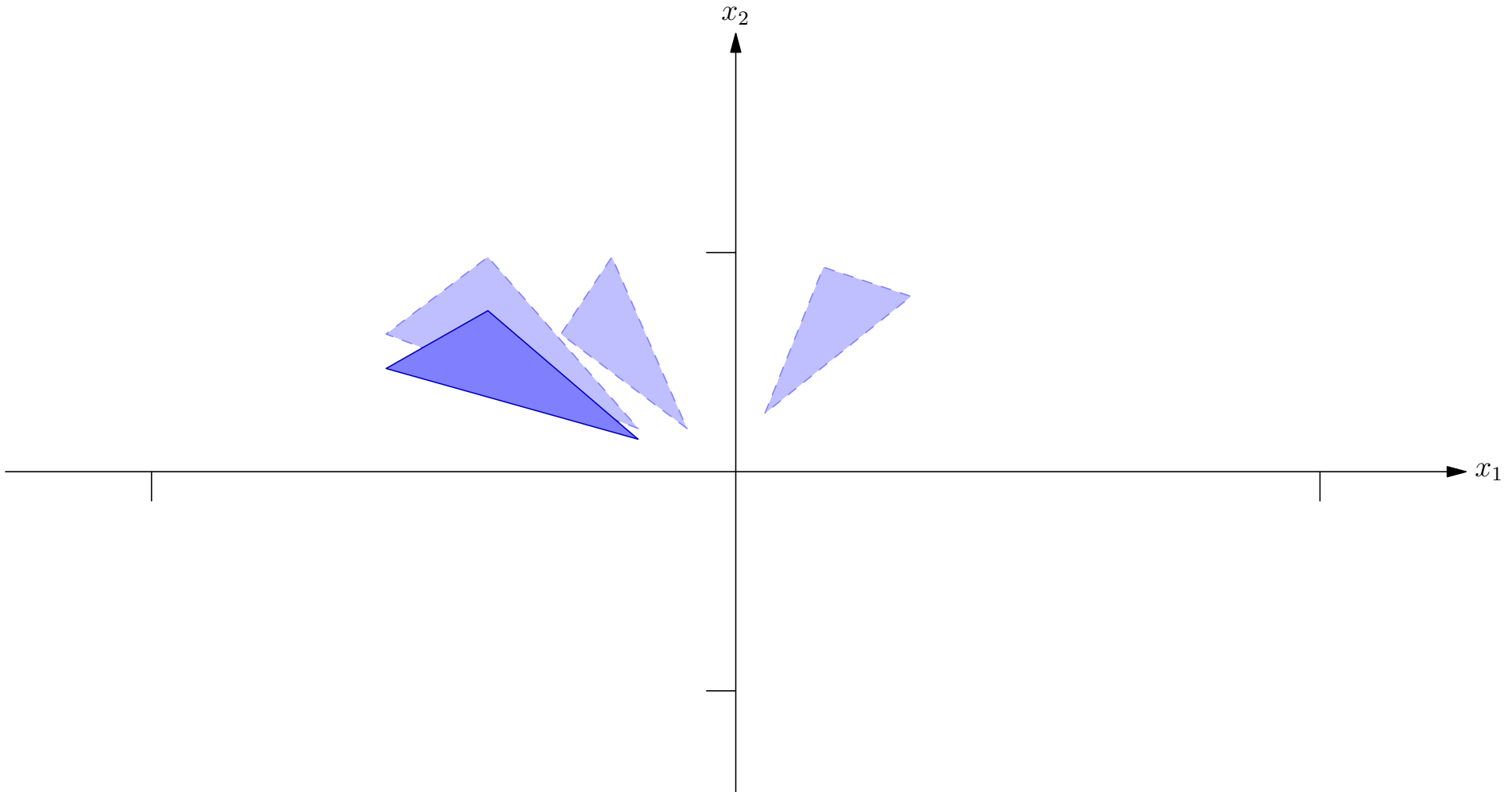
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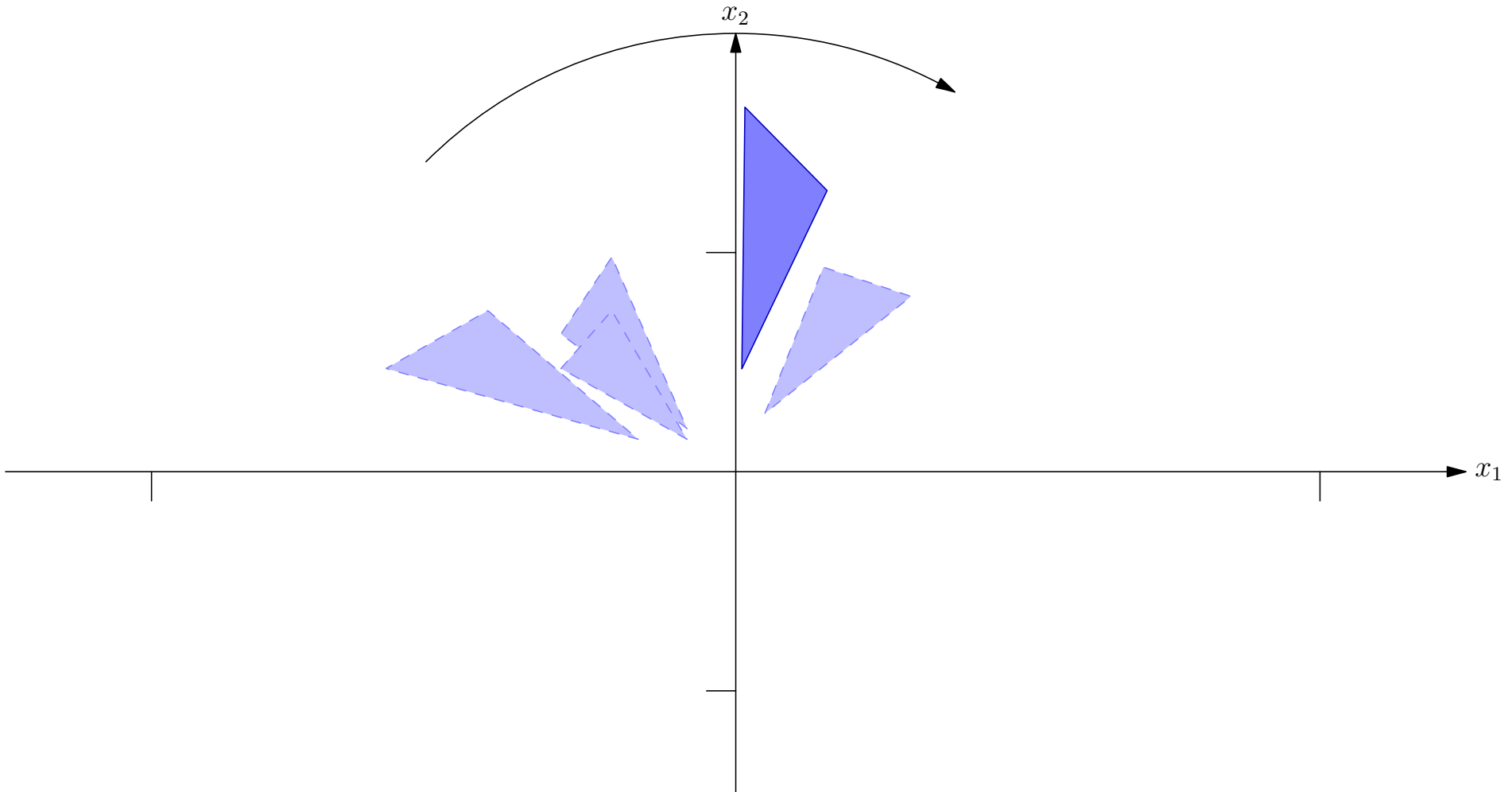
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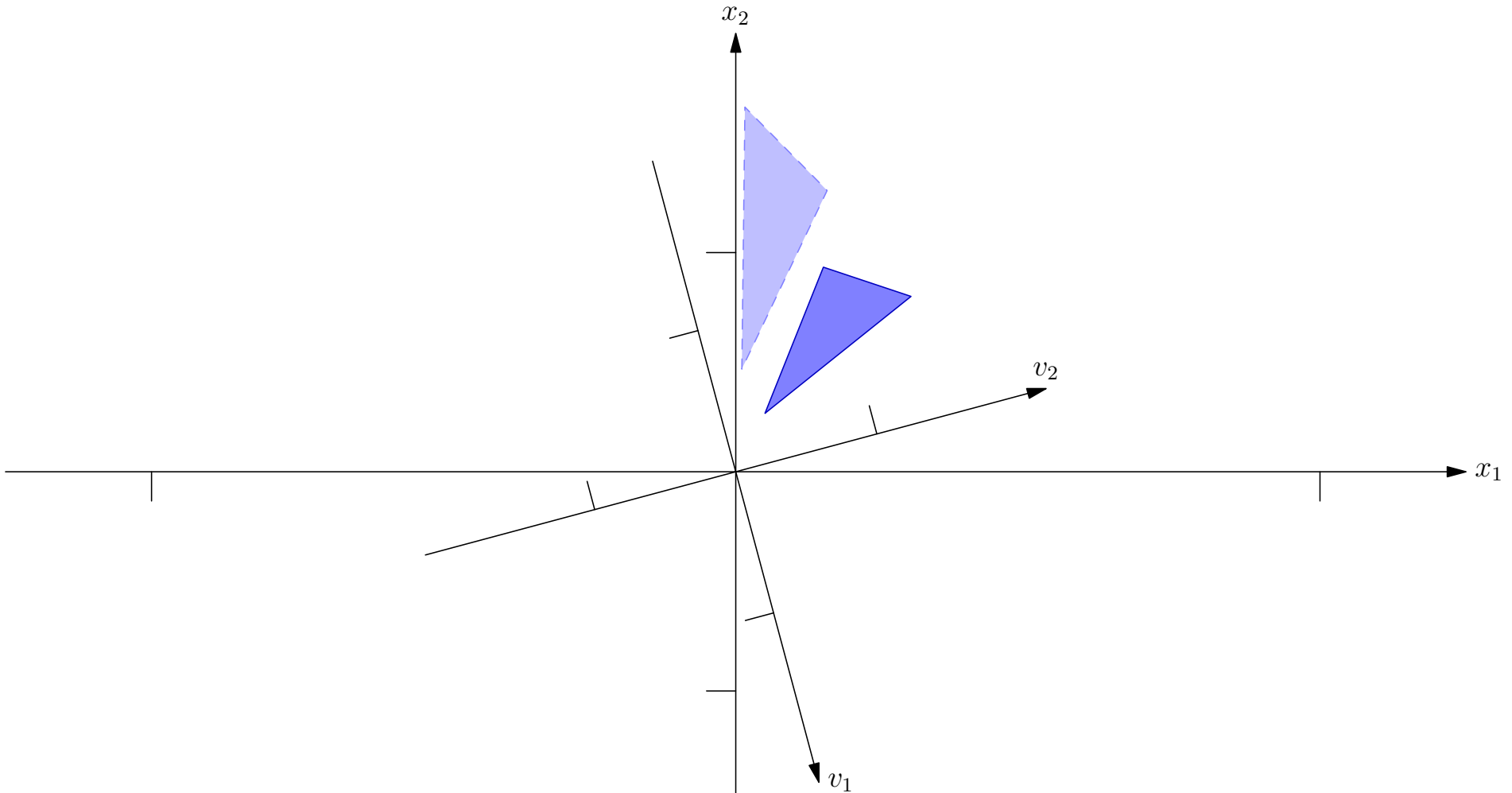
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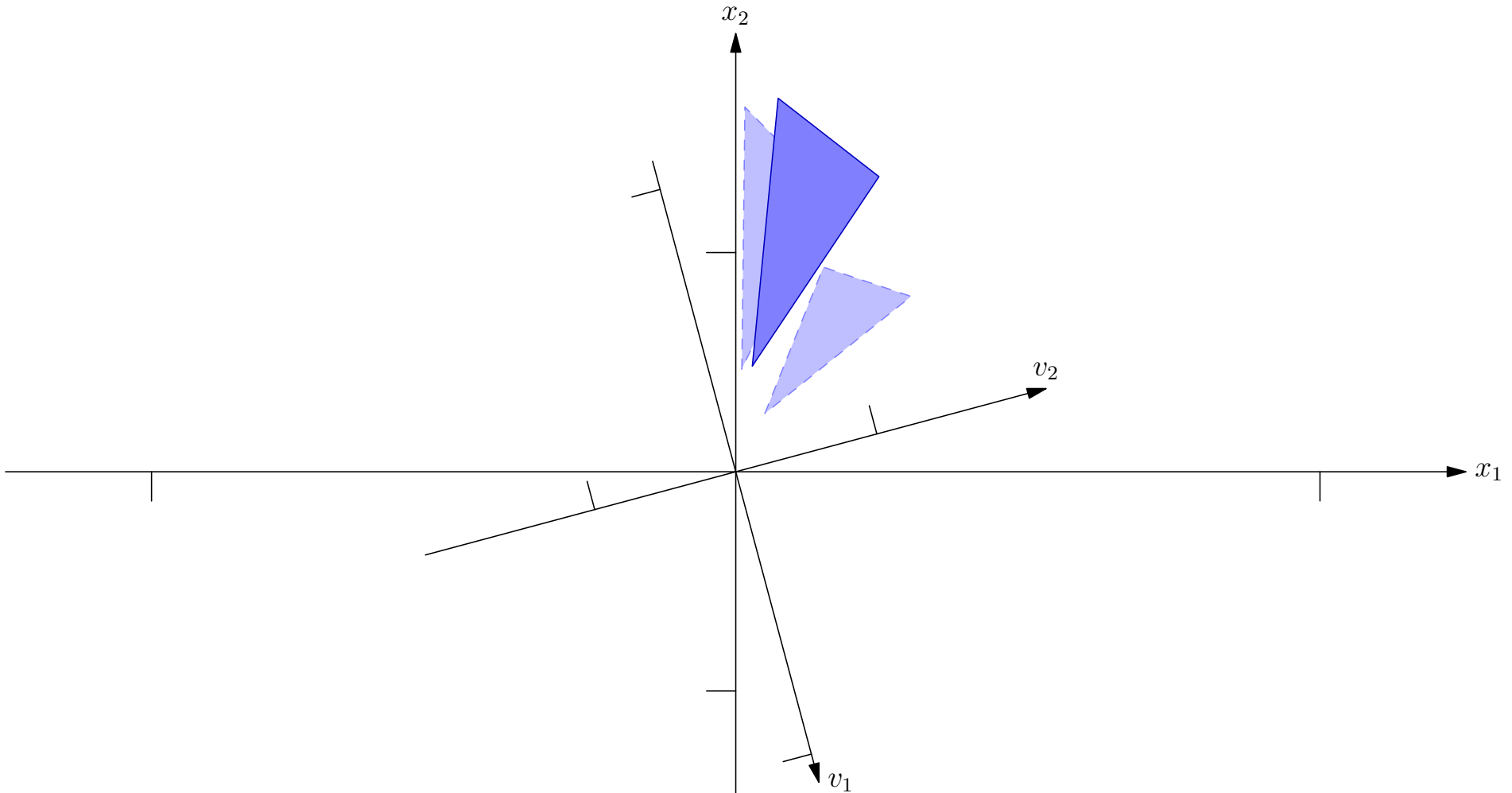
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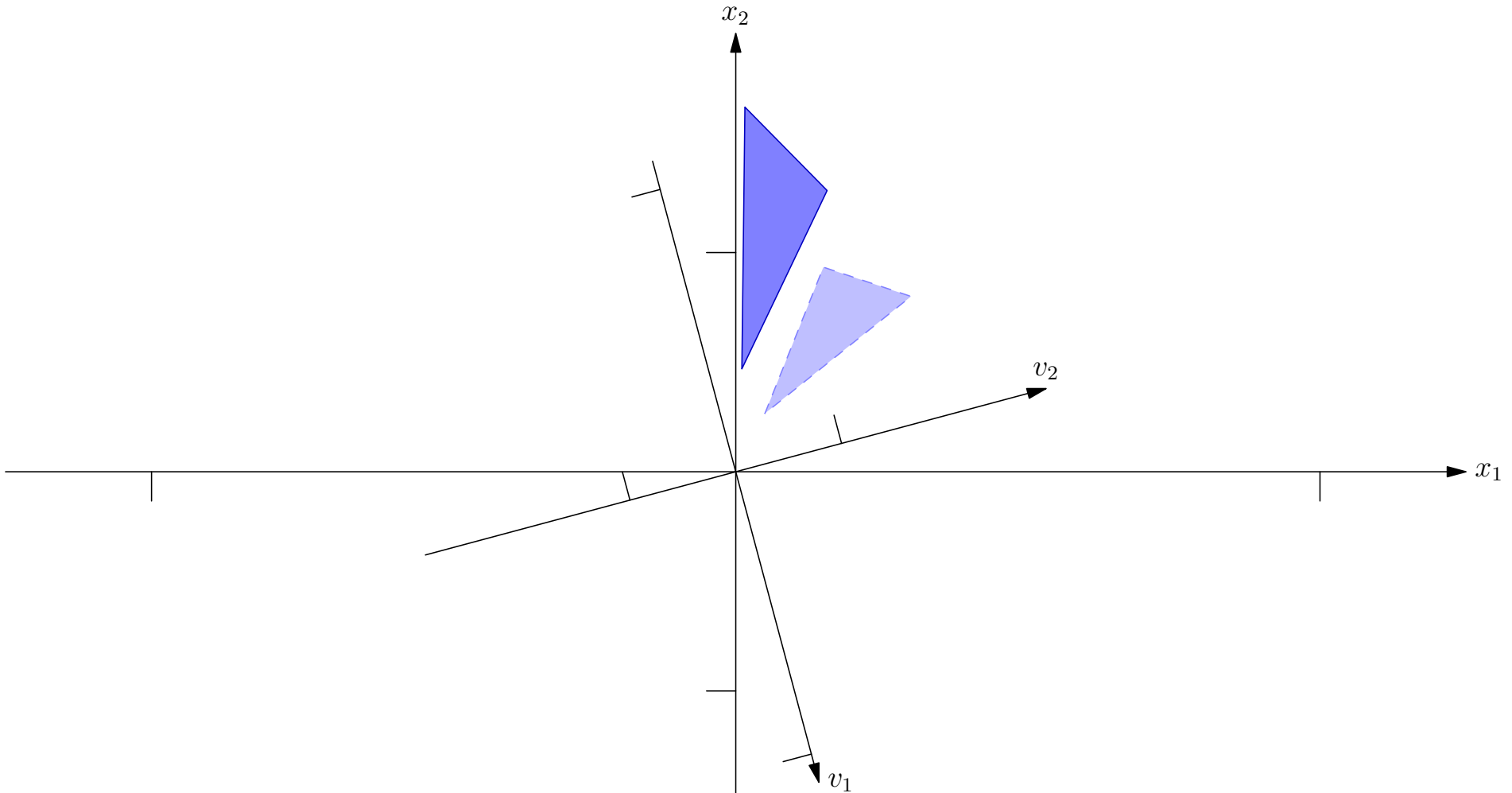
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# Inverses

- For any symmetric invertible matrix

$$\mathbf{M} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^T$$

$$\mathbf{M}^{-1} = \mathbf{V}\mathbf{\Lambda}^{-1}\mathbf{V}^T$$

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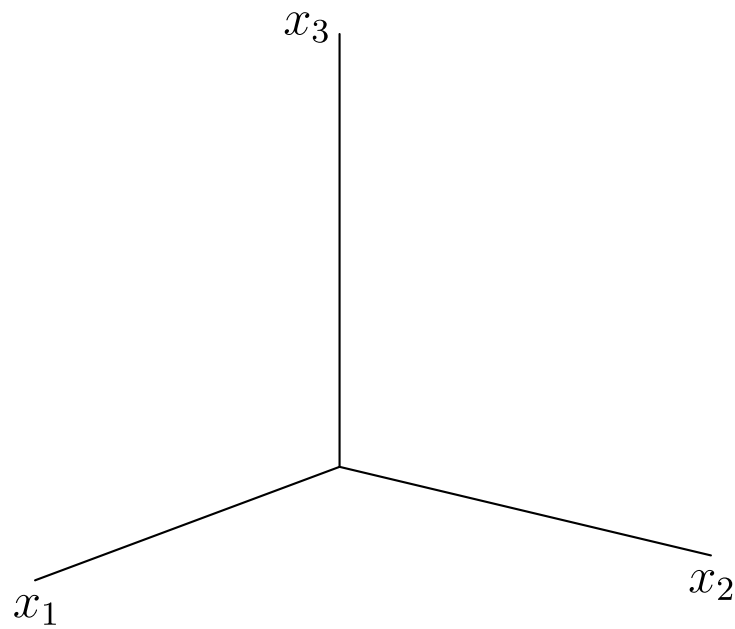
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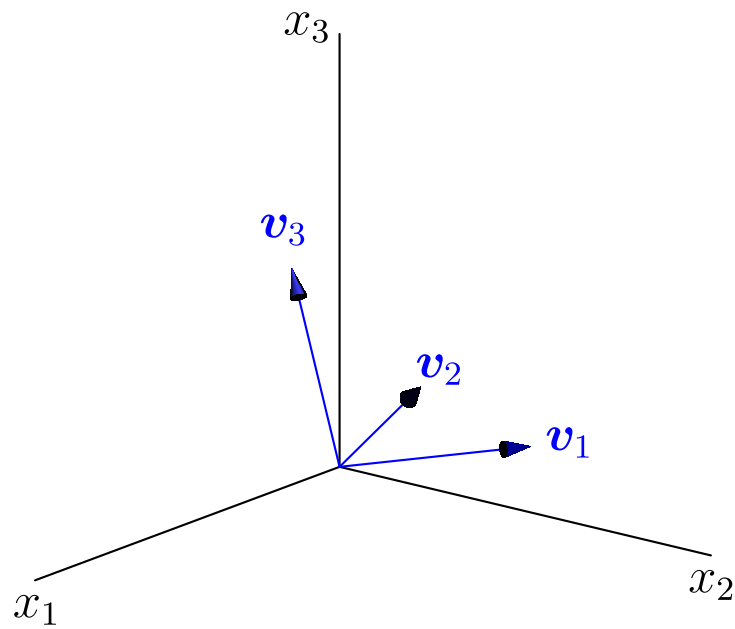
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# III-Conditioning Again



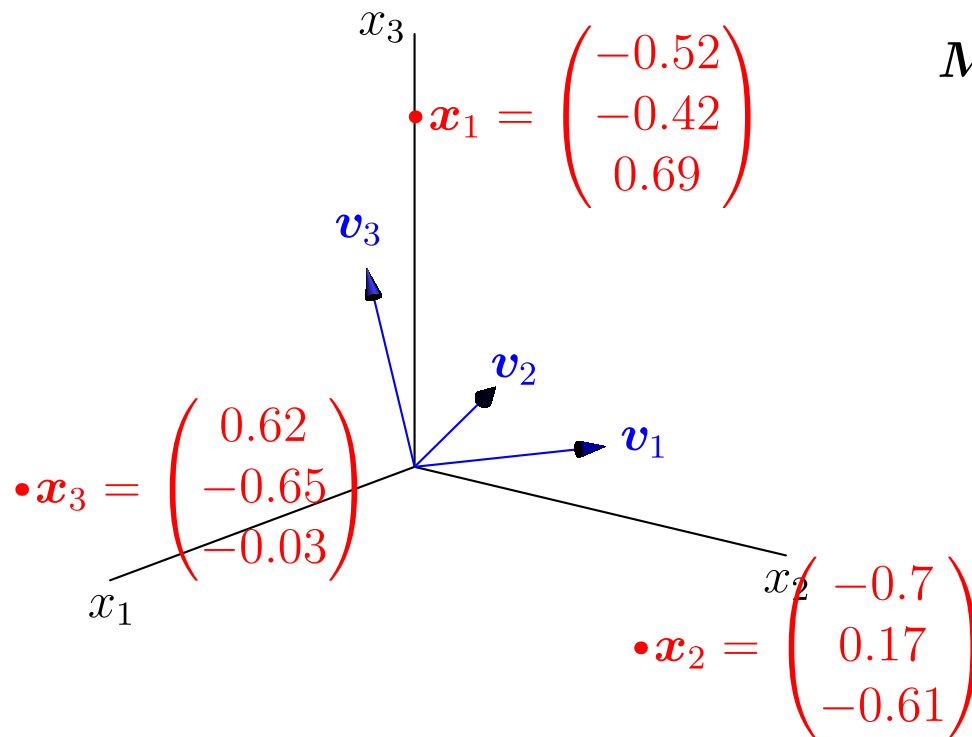
$$\begin{aligned} \mathbf{M} &= \begin{pmatrix} 0.80 & 0.01 & 0.25 \\ 0.01 & 0.85 & 0.48 \\ 0.25 & 0.48 & 0.5 \end{pmatrix} \\ &= \mathbf{V} \begin{pmatrix} 0.8 & 0 & 0 \\ 0 & 1.25 & 0 \\ 0 & 0 & 0.1 \end{pmatrix} \mathbf{V}^T \end{aligned}$$

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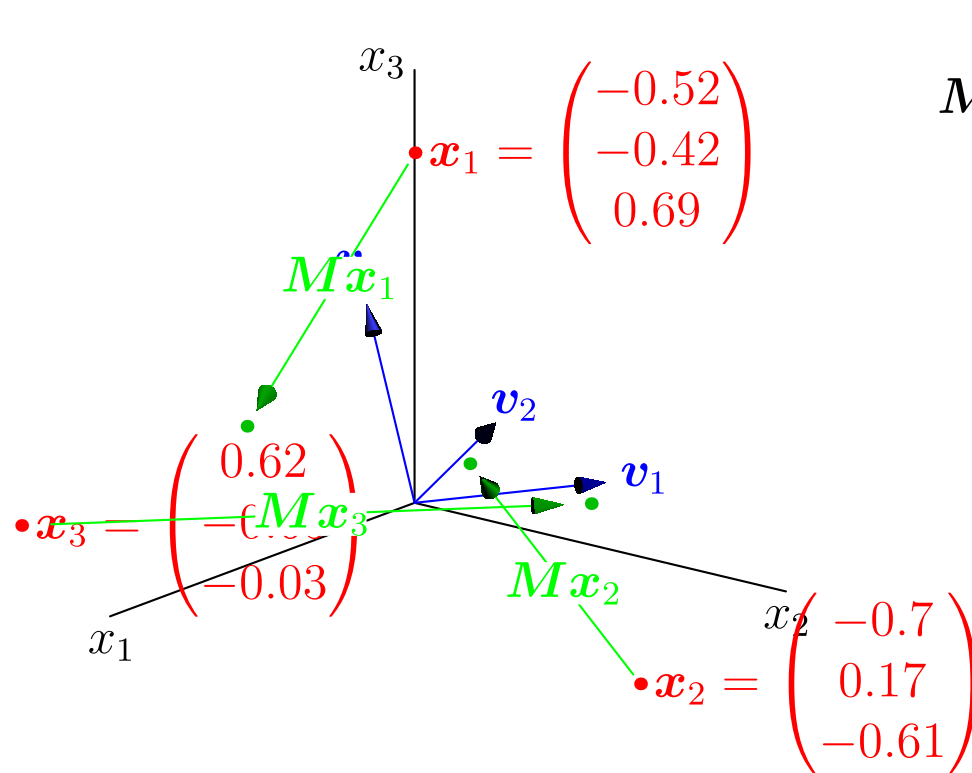
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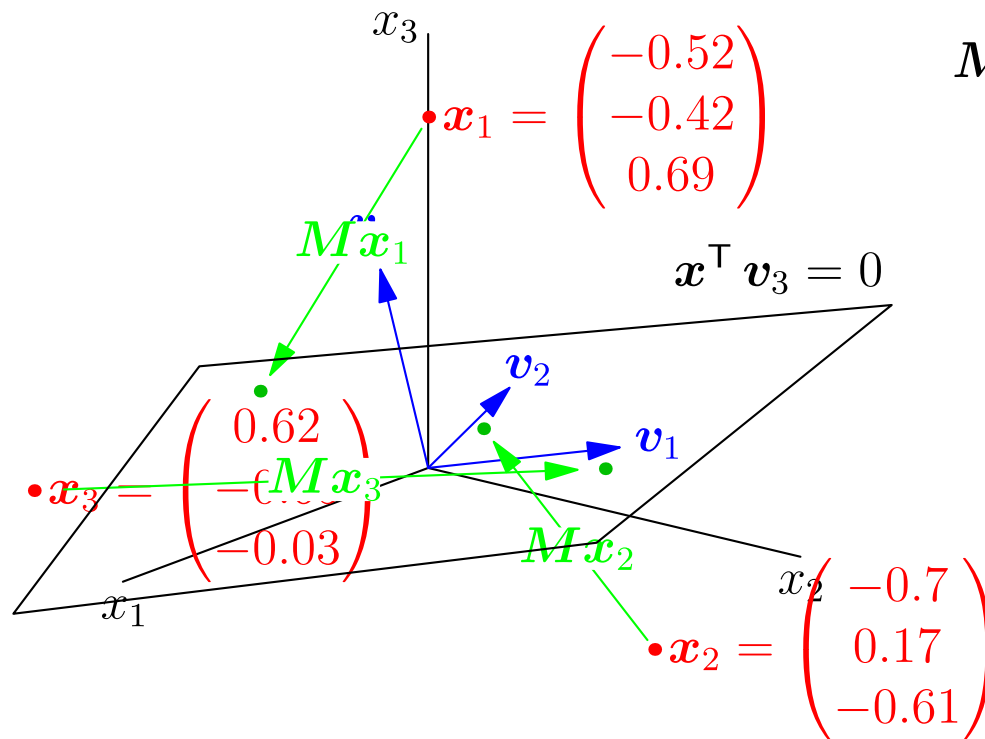
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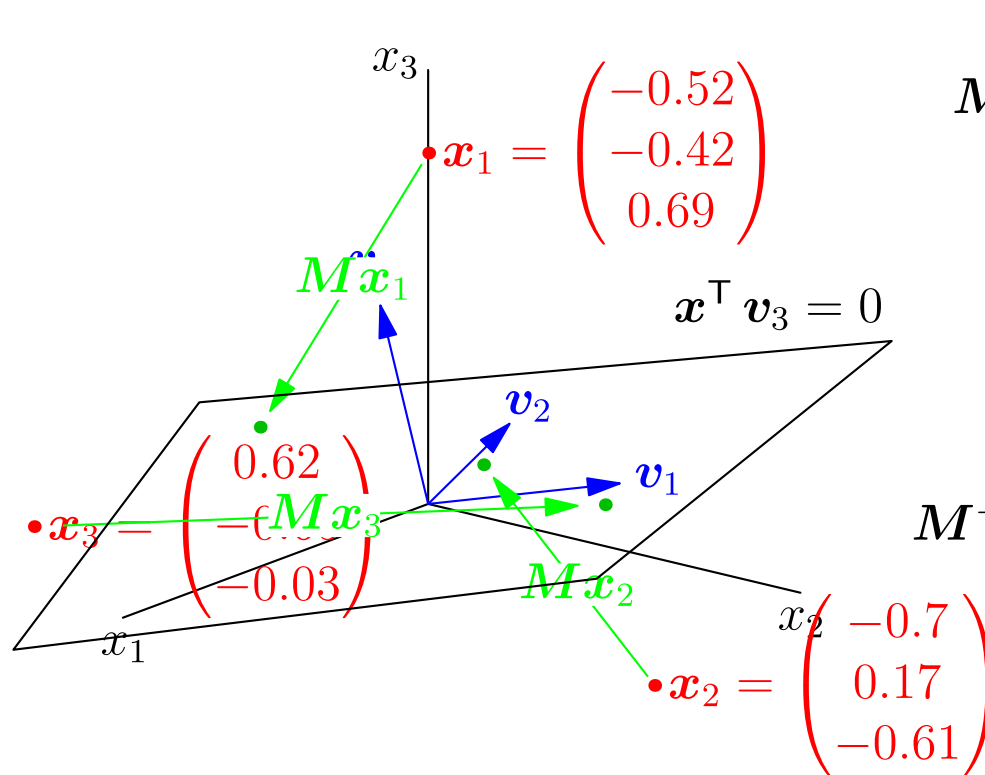
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$$M^{-1} = \begin{pmatrix} 1.9 & 1.17 & -2.1 \\ 1.17 & 3.34 & -3.8 \\ -2.1 & -3.8 & 6.8 \end{pmatrix}$$

$$= V \begin{pmatrix} 1.25 & 0 & 0 \\ 0 & 0.8 & 0 \\ 0 & 0 & 10 \end{pmatrix} V^T$$

# Condition Number

- Taking matrix inverses can be inherently unstable
- Any small error can be amplified by taking the inverse
- The stability of the inverse depends on the ratio of smallest eigenvalue to the largest eigenvalue (i.e. the biggest possible amplification compared to the smallest)
- For symmetric matrices the largest  $|\lambda|$  is the spectral norm,  $\|\mathbf{M}\|_2$ , from the vector spaces lecture
- The condition number is given by

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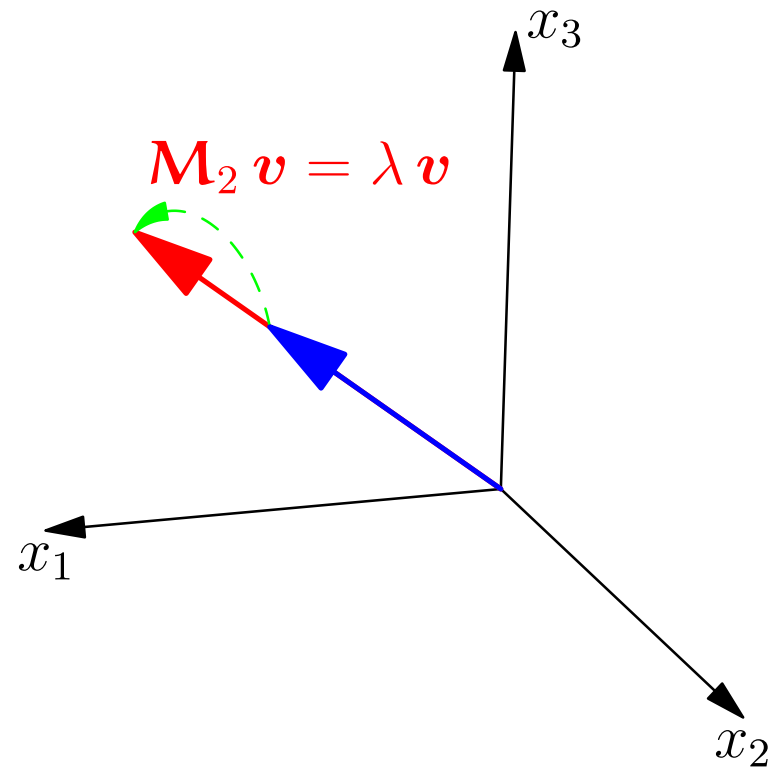
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# Outline

1. Eigenvectors
2. Orthogonal Matrices
3. Eigen Decomposition
4. **Low Rank Approximation**
5. Non-Normal Matrices



# Rank of a Matrix

- The rank of a symmetric matrix,  $\mathbf{M}$ , is the number of non-zero eigenvalues
- The space spanned by the eigenvectors  $\mathbf{v}_a$ ,  $\mathbf{v}_b$ , etc. with zero eigenvalue forms a **null space**
- Any vector in the null space will get mapped to the zero vector

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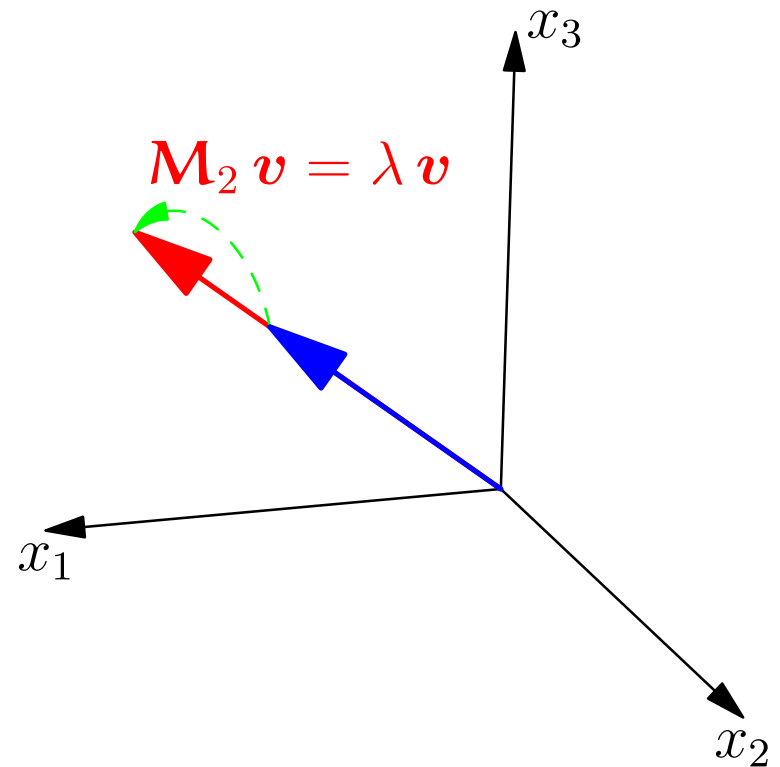
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# Non-Normal Matrices

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- Symmetric matrices are part of a class of matrices known as **normal matrices**, which satisfy  $\mathbf{M}\mathbf{M}^\dagger = \mathbf{M}^\dagger\mathbf{M}$  (for real matrices  $\mathbf{M}^\dagger = \mathbf{M}^\top$ )
- Hermitian matrices,  $\mathbf{H}$ , are complex matrices that equal their conjugate transpose:  $\mathbf{H} = (\mathbf{H}^\top)^* = \mathbf{H}^\dagger$
- Hermitian matrices are also normal; they are important in quantum theory, less so in machine learning
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# Asymmetric Matrices

- For  $n \times n$  real symmetric matrices we are guaranteed there are  $n$  orthogonal real eigenvectors
- For non-symmetric matrices the eigenvectors will generally not be orthogonal, they won't necessarily be real
- We are not even guaranteed there will be  $n$  linearly independent eigenvectors
- Matrices which don't have  $n$  linearly independent eigenvectors are said to be **defective**, e.g.  $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$  has only one
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# Characteristic Equation

- Since our eigenvectors are defined by  $\mathbf{M}\mathbf{v} = \lambda\mathbf{v}$  then

$(\mathbf{M} - \lambda\mathbf{I})\mathbf{v} = \mathbf{0}$  has a solution  $\mathbf{v} \neq \mathbf{0}$  only if  $\det(\mathbf{M} - \lambda\mathbf{I}) = 0$

- This is known as the characteristic equation and leads to an  $n^{th}$  order polynomial in  $\lambda$
- An  $n^{th}$  order polynomial has  $n$  roots (counting repeats), which are in general complex
- Thus even for real (non-symmetric) matrices we can get complex eigenvalues

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# Complex Conjugate Pairs

- If  $\mathbf{M}$  is real and  $\lambda$  is complex then  $\mathbf{v}$  must be complex too, otherwise  $\mathbf{M}\mathbf{v}$  would be real but  $\lambda\mathbf{v}$  would not
- Furthermore

$$\mathbf{M}\mathbf{v} = \lambda\mathbf{v} \quad \implies \quad \mathbf{M}\mathbf{v}^* = \lambda^*\mathbf{v}^*$$

- So that eigenvectors and their corresponding eigenvalues exist in complex conjugate pairs

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# Left and Right Eigenvectors

- For symmetric matrices if  $\mathbf{v}$  is a right-hand eigenvector in the sense that  $\mathbf{M}\mathbf{v} = \lambda\mathbf{v}$  then  $\mathbf{v}^\top$  is a left-hand eigenvector

$$\mathbf{v}^\top \mathbf{M} = \lambda \mathbf{v}^\top$$

- Earlier we used this property to show that the eigenvectors for symmetric matrices are orthogonal
- For asymmetric real matrices the left-hand eigenvectors  $\mathbf{u}^\top$  are different from the right-hand eigenvectors, although they share the same eigenvalues (they have the same characteristic equation)

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# Eigen-Decomposition

- For a non-symmetric matrix,  $\mathbf{M}$ , provided it is not defective, we can decompose the matrix as

$$\mathbf{M} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1}$$

- $\mathbf{\Lambda}$  is a diagonal matrix of eigenvalues
- $\mathbf{V}$  is a matrix whose columns are the right-hand eigenvectors
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# Summary

- Linear mappings are commonly used in machine learning algorithms such as regression
- We can understand symmetric operators by looking at their eigenvectors
- Any symmetric matrix can be decomposed as  $\mathbf{M} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^T$ 
  - ★ where  $\mathbf{V}$  is an orthogonal matrix whose columns are the eigenvectors
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- This decomposition allows us to understand inverse mappings
- Non-symmetric matrices are more complex

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- Non-symmetric matrices are more complex—we use the eigen-decomposition less because of this